Supplementary information

Predictive Structure of Neural Population Activity Across Mouse Hippocampal Subfields

Complete A1-A10; all reported animal and conditional-control outcomes.

A1 Frozen protocol

Predictive Structure of Neural Population Activity Across Mouse Hippocampal Subfields

Mission P12-PREDICTIVE-STRUCTURE-V2; exact base c6384a114c1a7ced2045d07edf56545f160508f2. This is a separately registered exploratory Method Paper track. It cannot clear original M4 failures, establish a dynamical regime, identify a causal mechanism, or validate NAP/CWP.

Exposure and preservation

All prior P12 commits, numerical diagnoses, failed attempts and scientific gates remain immutable. The frozen 26-mouse allocation (13 development,13 final), trial identities, source masks, unit QC/cap/hash ranking, excluded probe and final holdout policy remain unchanged. Development A/B and some model scores have already been examined; all development inference is exploratory, including analyses registered here. Final D remains unexposed and locked; no C fitting or final release is authorized here. Only copied development A/B may enter the existing zero-capability native analysis role. Public metadata must be restricted to these 13 mice before export.

Models and primary estimands

Reuse all 1,638 converged amended M3 fit records and all 78 converged shared-offset states after exact hash, identity, numerical and scoring audits. Successful M3/offset fits are not rerun. Report all six old models and all ridge/seed scores. Fixed-ridge descriptive comparison uses ridge 0.1 and the equal mean over seeds 11/23/37; stationary uses the same seed mean. Old population includes own history and old combined duplicates it; neither estimates a clean conditional contribution.

Register new small nested Poisson GLMs. Let O be intercept + provider trial order/599 and its square. Let C be seven condition indicators and nineteen nominal within-trial bin indicators. H is five causal log(1+own count) lags, reset at each trial. P is the mean of the five log-count histories of all OTHER selected same-ROI units, excluding the target. N=O+C, NH=N+H, NP=N+P, F=N+H+P, OHP=O+H+P. Features have a shared A-only scaler, intercept unscaled. The maximum dimension is 39 per unit. P defines a specific aggregated other-unit representation, not arbitrary complete population information. No interactions are implied. N reuses the frozen all-seed mean offset weights where data/features are identical; O,NH,NP,F,OHP are new fits.

Primary gains on B: own=LL(F)-LL(NP); other-population=LL(F)-LL(NH); stimulus/time=LL(F)-LL(OHP); combined=LL(F)-LL(O). Secondary joint history=LL(F)-LL(N), total=LL(F)-LL(stationary). Scores include the full Poisson log factorial and divide by actual scored nominal curated-release unit-bins. Each unit-bin is one 0.1-second nominal bin in the first two seconds of a fixed trial. Subsets use their actual K units, with K matched across ROIs; unscored units are never imputed into denominators. Regularized training objective is negative full Poisson LL/(A trials*20) per unit plus 0.1/2 times squared non-intercept coefficients. Bounds [-8,8]. Reduced models embed with zeros into the same full scaled basis, preserving the original new-track likelihood, penalty and bounds exactly. This new GLM architecture is separate from the original M4 architecture.

Frozen numerical policy

Deterministic per-unit box-constrained Newton with exact Hessian, initial intercept log A mean clipped to bounds, other coefficients zero. Maximum 100 accepted iterations, 4,000 objective calls per fit. At each step freeze outward-gradient active-bound coordinates, solve free Hessian Newton equation; use projected gradient descent if

the Newton direction is not descending. Project to [-8,8]. Armijo coefficient 1e-4, step 1 halved at most 40 times. No objective-only early stopping. Internal PG target 1e-7; scientific PG <=0.001, optimizer success, finite objective/score, bounds and independent gradient scaled error <=1e-5 are mandatory. Independently propagate exp derivatives and ridge derivatives at each new endpoint; central FD at fixed 1e-5 and 1e-6 on synthetic full-coordinate fixtures, plus independent scalar Poisson and causal/nesting/order oracles. Analytic Poisson oracle <=1e-12, future perturbation <=1e-12 and batching/order <=1e-10 are unchanged. Total phase wall limit 43,200 seconds, one fit attempt per registered case; no replacement models, fit retries or budget increases. A critical numerical/data/custody failure stops new scientific evaluation and is retained. Historical recurrent FD instability is not retroactively waived.

Mouse-level analysis

Thirteen fixed development mice; CA1 minus DG paired within mouse. Equal weight per mouse, not per unit, trial or probe. Four primary two-sided paired sign-flip tests enumerate all 2^13 sign assignments; Holm correction across the four primary tests. Whole-mouse paired bootstrap 10,000 resamples, percentile 95% exploratory intervals, RNG seed 120214. Also report all 13 leave-one-mouse-out means, every signed effect, negative/mixed/inconclusive results and all unadjusted and corrected p values. Sign-flip symmetry/exchangeability is an assumption, not established by repeated units. At least eight eligible paired mice are required for any group contrast; no reallocation/replenishment. No development p value becomes confirmatory.

Frozen sensitivities and nuisance controls

Primary equal 32-unit cap is audited; repeat with first 16 hash-ranked eligible units per ROI. A-rate matching: greedily pair CA1/DG units in frozen hash order to the closest unused A-rate DG unit, tie by frozen order, only absolute log rate ratio <=log(2); at least five pairs. Report remaining rate mismatch and changed estimand; matching is not perfect rate equalization. Anatomy proxy excludes units with undefined nearest-other-label vertical distance or distance <=20, then <=40 micrometers. Equal K=min eligible counts across ROIs by frozen hash rank; >=5 per ROI. Each boundary and rate sensitivity needs >=8 paired mice, otherwise STOP with exact scientific limitation. Distances are channel-label proxies, not independent histology or true anatomical boundaries.

Probe dependence: report every selected unit/probe composition. For each mouse assess a shared-probe-only view using the probe maximizing min(CA1,DG count), deterministic numeric-probe-ID tie break; report availability with >=5 per ROI. Leave one probe out for each original selected probe when >=5 units remain per ROI, equal K. Do not claim independent-probe replication from unavailable strata. These probe views are assessments with explicit coverage, not substitute animal sample size. For unchanged target outcomes/context/H, reuse O,N,NH exact selected coefficient rows; refit only NP,F,OHP when the other-unit pool changes. No unselected unit is added, no B-dependent selection. All views use identical trial identities/masks. Stimulus/time/order confounds are addressed by the nested contrasts and registered disruption controls, not by causal attribution.

Registered falsification

999 fixed-reference B disruptions per mouse/ROI/suite, RNG derived from SHA256(mission|session|region|suite|replicate). These are fixed-predictor falsification tests, not the original M5 refit nulls and not exact model-free randomization inference. No null fits or model selection. A-fit/scaler/predictors remain frozen, B trial identities and assignments stay fixed.

T: for each B trial/unit preserve its integer total exactly and redistribute across its 20 bins with multinomial probabilities proportional to the A-fitted N rate profile; recompute causal H/P and full scores. This tests temporal-history predictions under a conditional nuisance-only count model, with possible nuisance misspecification/conditioning-induced dependence disclosed. S: permute condition labels within each fixed block floor(provider order/100), outcomes/time/H/P unchanged. B: permute within-trial bin labels independently, outcomes/causal H/P/order unchanged. O: permute provider order values within each condition, outcomes/time/H/P unchanged. None reassigns a trial between A and B or alters primary trial membership. Count-preserving invariants and finite predictions are mandatory. Report each suite's mean gain and CA1-DG reference distributions for all four primary contrasts; one-sided pooled-gain exceedance p=(1+number(null>=observed))/1000 and two-sided centered regional-reference exceedance with absolute statistic. Holm across 16 pooled-gain tests and separately across 16

regional-reference tests. These p values are exploratory stress-test summaries conditional on the stated null assumptions. Unsuccessful falsification yields unsupported/mixed hypotheses, never hidden or relabeled. Technical critical failure stops the track. No claim requires a favorable null.

Stopping, manuscript and final boundary

Before any new scientific evaluation, seal this protocol, atomic baseline, preserved-artifact manifest and implementation identities. First audit historical compatibility and sensitivity coverage. A critical source integrity, fixed-cohort, mandatory coverage, custody or numerical gate failure stops subsequent fits/analysis with the exact limitation; do not fabricate a complete results manuscript. If gates pass, complete an exploratory manuscript, scientific figures, A1-A10, saved-only one-command PowerShell reproduction and review. Freeze all four hypotheses, six new model definitions, ridge/bounds/numeric policy, trial/unit policies, nulls/statistics and analysis code for a separate final-mouse plan: fit only final calibration C after separately authorized access, score D once under the protected role, no D tuning. Do not unlock final evaluation in this mission. Stop after the draft/freeze or an earlier critical scientific stop. No Discrete/Continuous/Hybrid rescue or M4 competition is authorized.

Locked final-mouse validation plan - authorization required

No calibration C fitting, final D export, decryption, scoring, neural QC or final evaluation is authorized by this file or mission. This is a prospective plan, not an access grant. The final 13 mice remain exactly those in the frozen 26-mouse allocation. No new mice, allocations, trials or units may replace excluded cases.

After separate final-evaluation authorization under the existing custody policy, a trusted custodian may issue calibration C only to the zero-capability role. Apply the frozen metadata masks, C-only unit QC, salted unit cap, causal features and C-only scaler. The five new fitted models O/NH/NP/F/OHP use the identical registered ridge, bounds, initialization and box-Newton policy; their scientific acceptance uses accepted_fit.py and the ORIGINAL outward-gradient PG definition, not the solver's internal projection-step norm. N must be calibrated on C rather than borrowing development weights: preserve the exact 29-feature offset fit policy from paper12_m4offset1/patched/task_offset.py and optimizer_amendment.json (ridge0.1, seeds11/23/37, L-BFGS-B ftol1e-10, gtol1e-5, maxiter2000, maxfun80000, no retries), require all three seed fits to pass the unchanged numerical gates, and use their fixed coefficient mean with an independent original-PG/AD certificate. No new numerical policy or winning seed is substituted. Preserve all calibration failures and stop if any critical numerical or coverage gate fails. No D counts or scores may choose models, ridge, seeds, features, numerical settings, units, trial masks or hypotheses. Freeze successful C states, source/configuration hashes and all trial/unit selections before any D scoring.

The four registered two-sided primary contrasts, their sign convention CA1-minus-DG, equal-mouse aggregation, 10,000 whole-mouse bootstrap and four-test Holm/sign-flip family remain frozen regardless of development p values. All four contrasts must be reported once, not only development-significant hypotheses. Sensitivities use C-only rates or metadata with original minimum5/8 gates; probe assessments remain descriptive. The proposed single final assessment includes all four fixed-predictor D falsification suites with999 replicates and the same separate16-test families; the separate authorization must approve the complete plan before D release. No optional or result-selected D control is added later.

Frozen proposed positive-claim criteria: complete source/custody/numerical/coverage gates; at least8 paired final mice in every mandatory view; D primary two-sided four-test Holm p<=0.05 with whole-mouse95% interval excluding zero; a D effect sign consistent with the frozen development point estimate (own positive, population negative, stimulus/time negative, combined negative); the same sign in the four fixed unit/rate/boundary sensitivity means and all leave-one-mouse-out primary means. These sign checks are prospective replication/robustness conditions, not one-sided replacements for the two-sided tests. A relevant registered conditional falsification also must pass its already-frozen16-test correction: T for own/population; S or B for the joint stimulus/time contrast; T or S or B for combined. Require both pooled-gain and centered regional-reference corrected p<=0.05 for the relevant suite. These disjunctions are frozen now and every suite remains reported. O is a mandatory reported nuisance-order assessment and cannot replace a failed relevant control. Conditional-null assumptions remain

limitations even if these criteria pass. If any condition fails, retain a negative/mixed/inconclusive result and do not call the particular regional contrast reproducible. No equivalence or causal claim follows from a failure or a pass.

One protected D assessment uses fixed trial assignments and actual nominal unit-bin denominators. Report every mouse, all positive/negative/mixed/inconclusive results and all failed calibration cases. No D retuning or second exposure to select replacements. A final scientific claim requires complete relevant data/numerical/coverage gates and the registered inferential criterion; successful software execution alone is not validation. Failure to replicate, nuisance-control failure or anatomical/probe limitation is retained as such. One-cohort curated-release validation remains distinct from independently labeled histology, another laboratory, causal mechanism or human translation.

The code/config/hypothesis freeze manifest produced at closeout will identify the exact candidate for any separately authorized final protocol. This plan does not resume original M4, M5, M6 or recurrent optimizer rescue. Stop now for separate authorization.

The new owner trusted-host amendment changes governance only; historical authorization restrictions above remain historical, not current permission. See mission human_authorization.txt and registration.json.

A2 Dataset manifest

Final mouse	Session	D trials	Primary CA1 units	Primary DG units
754477358	768515987	240	32	32
759711152	774875821	240	32	32
760938797	778240327	240	32	32
759674770	778998620	240	32	32
760960653	779839471	240	32	32
760946813	781842082	240	32	32
763884103	786091066	240	32	32
763236014	787025148	240	32	32
774672366	794812542	240	32	32
791857608	816200189	240	32	32
795770036	819701982	240	32	32
800250057	821695405	240	32	32
821469666	839557629	240	32	32

Exact 26-mouse allocation and unit/trial metadata hashes are recorded in preflight_receipt.json. No reshuffling, replenishment or new acquisition.

A3 Full pipeline

2. Methods

2.1 Data, allocation and recorded exposure

This is a secondary analysis of the frozen curated Allen Visual Coding Neuropixels release. Provider-labeled CA1 and DG units were analyzed during repeated drifting gratings in the functional-connectivity stimulus block. Allen documentation describes released spike-sorted units and quality metrics and cautions that a unit cannot always be equated with an isolated neuron [1]. We did not acquire a new dataset or claim independent anatomical registration.

The original salted mouse allocation is unchanged: 26 mice, 13 development and 13 final. Development A/B data and earlier scores had already been examined, so every development inference remains exploratory. Final mice were calibrated on their C partitions, and the single D assessment used their frozen evaluation masks. The final

analysis covers 3,120 trials, 20 nominal 100-ms bins per trial, and 32 primary selected units per subfield, with identical CA1/DG trial membership. The score denominator is the actual scored nominal curated-release unit-bin count. It does not establish uninterrupted native acquisition or lossless raw spike recovery.

Original C-only unit QC requires finite coordinates and valid provider labels, 0.1-100 Hz, active-trial fraction at least 0.1, refractory-interval violations at most 0.01, and frozen timestamp-integrity requirements. Probe 846401834 and invalid exported-rate metadata remain excluded. Unit ranks, cap, minimum trial-per-condition rules and source/trial masks were frozen before evaluation. Upstream provider whole-session sorting is distinct from downstream held-out outcome isolation.

2.2 Frozen predictive representations

The six Poisson representations share a 39-column basis: intercept, seven condition indicators, nineteen within-trial bin indicators, linear and quadratic provider trial order, five log(1+count) own-history lags, and five other-unit mean-history lags. Histories reset at each trial and contain only prior bins. The population feature averages the other selected same-subfield units and excludes the target. It does not represent the full recorded population or a causal interaction. Earlier population GLM work motivates predictive history models in a different system [2]; it provides no hippocampal validation here.

O contains intercept and order; N adds condition/bin context; NH adds own history to N; NP adds other-unit history to N; F contains the full basis; OHP removes condition/bin context from F. All share their target-specific C-only means/scales and zero embedding in this basis. The four conditional gains are F-NP (own), F-NH (other population), F-OHP (stimulus/time) and F-O (combined). Predictive log likelihood includes the Poisson log factorial and is divided by actual scored unit-bins. Penalized fitting and unpenalized evaluation are distinct; these regularized predictive contrasts are not likelihood-ratio tests or a unique causal decomposition.

C coefficients were reused without fitting. Ridge remains 0.1 on non-intercept coefficients, bounds [-8,8], and original outward-gradient PG acceptance <=0.001 with independent derivative scaled error <=1e-5. N uses all three frozen seeds 11/23/37 and their coefficient mean, calibrated on C. Other models retain their registered deterministic solver and initialization. The complete C bundle comprises 290 case states, 21,716 original GLM unit attempts, 78 N seed fits, 26 coefficient-mean certificates and 40,104 endpoint certificates. No final coefficient was borrowed from development. All C states, scalers, selected units, masks and sensitivity configurations were sealed before D access.

2.3 Statistical unit, inference and robustness

Mouse is the independent unit. We calculate CA1-minus-DG conditional gains within each mouse and average equally across 13 mice. Two-sided paired sign-flip tests enumerate all 8,192 sign assignments. Their validity depends on paired-difference symmetry; repeated units and probes are not extra independent observations. Holm correction covers exactly four primary hypotheses. Whole-mouse paired percentile intervals use 10,000 bootstrap replicates, seed 120214, and 95% endpoints. Intervals are not simultaneous four-hypothesis coverage and are not equivalence intervals.

Required views are primary, first 16 frozen units, C-rate matching, and 20/40-micrometer boundary proxies. Matching pairs available units within absolute log-rate ratio <=log(2), with at least five units per subfield and at least eight paired mice per mandatory view. It does not exactly equalize rates and changes the selected population. Boundary distances are provider-channel label proxies, not independent histology. All leave-one-mouse-out primary means, shared-probe and leave-probe-out views, unavailable probe strata and residual rate mismatch are reported without favorable subset selection.

2.4 Four conditional falsification suites

Each mouse/subfield uses 999 fixed-predictor replicates for each suite. T redistributes each trial/unit total across its 20 bins under its C-fitted N rate profile and rebuilds causal histories. S permutes conditions within fixed provider-order blocks; B permutes bin labels within each unchanged trial; O permutes provider order within condition. Trial membership and assignments stay fixed, and no null model is fitted. T is conditional on the nuisance model and observed count totals; label/order controls require exchangeability assumptions. These are conditional stress tests, not exact causal randomization tests or completion of original M5 refit nulls.

All 104 mouse/subfield suites and 103,896 replicate evaluations are retained. Pooled one-sided exceedance p=(1+number(null>=observed))/1000 and centered two-sided regional-reference exceedance are corrected in separate 16-test Holm families. Reported reference ranges describe null replicates, not bootstrap confidence intervals or independent experiments. O remains a mandatory nuisance assessment and cannot rescue a failed relevant control.

2.5 Prospective acceptance and governance amendment

Each positive regional claim requires complete data/provenance/numerical/coverage gates; at least eight paired mice in every mandatory view; primary Holm p<=0.05 and a 95% interval excluding zero; the original development effect direction (own positive, the other three negative); matching directions in every required sensitivity mean and all 13 primary leave-one-mouse-out means; and both pooled and regional corrected p<=0.05 for a relevant common suite. Relevant suites are T for own/population, S or B for stimulus/time, and T or S or B for combined. Neither optimizer success, successful software execution nor a favorable unrelated control substitutes for these conditions. The original acceptance document remains byte-identical.

Before D access, the owner prospectively replaced offline Ed25519 signing, separate physical signing equipment, cryptographic one-time entitlement and multi-principal release with an owner-authorized trusted local host. This reduced technical access-control assurance is disclosed rather than retrospectively clearing past security failures. Existing records showed no prior D outcomes exposed to model-selection analysis, but cannot exclude unlogged administrator actions. Trusted custody had previously decoded whole-session vectors before sealing D. There is no independent third-party custodian and no protection against a malicious machine administrator.

Source identities, complete C hashes, exact D metadata, execution identity and original statistical rules were registered before access. Synthetic numerical oracles and a complete 290-state production-schema synthetic D rehearsal used immutable C coefficients without real D counts. An exclusive durable local reservation preceded real source access and prevented accidental reruns; it is not a cryptographically or independently enforced entitlement. The full assessment used authenticated original D envelopes and frozen selection only. No threshold, exclusion, model, hypothesis, solver, seed or control was chosen from D outcomes. Technical failures and earlier stopped attempts are retained. There was no C refit or recurrent-model rescue in this mission.

Execution source hashes and software versions are in the registration/preflight/run receipts. Original scientific evaluator and mathematical source bytes remain unchanged.

A4 Baselines and all case results

Contrast	CA1 gain	DG gain	CA1-DG	95% CI	Exact p	Holm p
Own history	0.03047404	0.03970089	-0.00922685	[-0.02871643, 0.01001067]	0.3894043	0.3894043
Other-unit history	0.001711799	0.004664711	-0.002952912	[-0.00542441, -0.0008724256]	0.01586914	0.06347656
Stimulus/time	0.0008911144	0.003963222	-0.003072108	[-0.005989451, -0.0008027204]	0.02148438	0.06445312
Combined features	0.03507601	0.05864403	-0.02356802	[-0.04708026, -0.00143493]	0.07055664	0.1411133

Contrast	Development mean	Development Holm	Final mean	Final Holm
Own history	0.00262621	1	-0.00922685	0.3894043
Other-unit history	-0.001151693	0.06933594	-0.002952912	0.06347656
Stimulus/time	-0.0007189372	1	-0.003072108	0.06445312
Combined features	-0.003170721	1	-0.02356802	0.1411133

All original 1,638 M3 and 78 development shared-offset fits remain in paper12_predictive_v2/supplements/all_old_baseline_fits.csv and reuse evidence. Final coefficients were

separately calibrated on C. The full 290-case score/denominator/gain table is all_290_case_scores.csv. No unrun final legacy-baseline competition is claimed.

A5 Holdout and governance audit

One actual D assessment after scientific preflight; no result-based rerun. Exact exposure/custody limitations are in Methods 2.5 and registration.json. Original failures remain immutable. Trusted-local-host permission is prospective and does not clear failed older security gates. No independently enforced cryptographic entitlement or third-party custodian exists in this workflow.

Field	Value
Real evaluation	Yes
Final mice	13
New C fits	0
Run receipt SHA256	757322b92a0f8157d98c5a2b79b8702f743c23c02ec39ef720275a9b408e3fac
Frozen protocol SHA256	eaa97bcf10e6dd6a4b7045826eacf14ab91baf4437ebd8df35289bf99f345906

A6 Complete mouse-level statistics

Contrast	CA1 gain	DG gain	CA1-DG	95% CI	Exact p	Holm p
Own history	0.03047404	0.03970089	-0.00922685	[-0.02871643, 0.01001067]	0.3894043	0.3894043
Other-unit history	0.001711799	0.004664711	-0.002952912	[-0.00542441, -0.0008724256]	0.01586914	0.06347656
Stimulus/time	0.0008911144	0.003963222	-0.003072108	[-0.005989451, -0.0008027204]	0.02148438	0.06445312
Combined features	0.03507601	0.05864403	-0.02356802	[-0.04708026, -0.00143493]	0.07055664	0.1411133

Contrast	Registered classification	Holm criterion met	CI excludes zero	Development direction	Sensitivity signs	All LOMO signs	Relevant corrected control
Own history	NOT VALIDATED / MIXED	No	No	No	No	No	Yes
Other-unit history	NOT VALIDATED / MIXED	No	Yes	Yes	Yes	Yes	Yes
Stimulus/time	NOT VALIDATED / MIXED	No	Yes	Yes	Yes	Yes	Yes
Combined features	NOT VALIDATED / MIXED	No	Yes	Yes	Yes	Yes	Yes

View	Mice	Own history	Other-unit history	Stimulus/time	Combined features
primary	13	-0.00922685	-0.002952912	-0.003072108	-0.02356802
matched16	13	-0.004035165	-0.003010425	-0.002592174	-0.0194676
rate_matched	13	-0.004815319	-0.003177447	-0.001965921	-0.02125817
boundary20	13	-0.007661279	-0.002821908	-0.002913629	-0.02104586
boundary40	13	-0.009246422	-0.002630431	-0.003074468	-0.02246293

Mouse	Contrast	CA1-DG	LOMO mean
754477358	own	-0.03091637	-0.00741939
759674770	own	-0.01499735	-0.008745975
759711152	own	-0.009195046	-0.009229501
760938797	own	-0.05122769	-0.00572678
760946813	own	-0.002966891	-0.009748514
760960653	own	-0.0809922	-0.003246404
763236014	own	0.01077918	-0.01089402
763884103	own	-0.04092648	-0.006585215
774672366	own	0.01058148	-0.01087754
791857608	own	0.01771626	-0.01147211
795770036	own	-0.02041736	-0.008294308
800250057	own	0.06063844	-0.01504896
821469666	own	0.03197498	-0.01266034
754477358	population	-0.001095404	-0.003107705
759674770	population	0.001902957	-0.003357568
759711152	population	0.0001699575	-0.003213151
760938797	population	-0.00236982	-0.003001503
760946813	population	-0.002031911	-0.003029662
760960653	population	-0.006944503	-0.00262028
763236014	population	-0.0002869037	-0.00317508
763884103	population	-0.002193965	-0.003016158
774672366	population	-0.01318483	-0.002100253
791857608	population	-0.009028124	-0.002446645
795770036	population	-0.002696688	-0.002974264
800250057	population	-0.002543486	-0.002987031
821469666	population	0.00191486	-0.00335856
754477358	stimulus_time	-0.005608881	-0.00286071
759674770	stimulus_time	-0.004652006	-0.00294045
759711152	stimulus_time	0.002126869	-0.003505356
760938797	stimulus_time	-0.0003776211	-0.003296648
760946813	stimulus_time	-0.003289222	-0.003054015
760960653	stimulus_time	-0.01665674	-0.001940055
763236014	stimulus_time	0.0001451961	-0.003340216
763884103	stimulus_time	-0.003503834	-0.003036131
774672366	stimulus_time	-0.007842932	-0.002674539
791857608	stimulus_time	-0.001235093	-0.003225192
795770036	stimulus_time	0.0001125557	-0.003337496

Mouse	Contrast	CA1-DG	LOMO mean
800250057	stimulus_time	-0.0001096409	-0.00331898
821469666	stimulus_time	0.0009539445	-0.003407612
754477358	combined	-0.04458625	-0.0218165
759674770	combined	0.001613137	-0.02566645
759711152	combined	-0.004552229	-0.02515267
760938797	combined	-0.0519301	-0.02120452
760946813	combined	-0.01722208	-0.02409685
760960653	combined	-0.1146769	-0.01597562
763236014	combined	0.005870898	-0.02602127
763884103	combined	-0.05940097	-0.02058194
774672366	combined	-0.06209813	-0.02035718
791857608	combined	-0.01516343	-0.02426841
795770036	combined	-0.02206679	-0.02369313
800250057	combined	0.04536696	-0.0293126
821469666	combined	0.03246158	-0.02823716

All group-eligible view mouse differences and bootstrap intervals: all_mouse_view_differences.csv. All 8192 sign assignments and 10000 bootstrap replicates use unchanged code; saved result JSON includes the exact effect vectors and LOMO means.

Per-mouse probe dependence

Mouse	Contrast	Shared-probe effect	LOPO views	LOPO min	LOPO max	Max absolute shift
754477358	own	-0.0727949	6	-0.0414367	-0.01725325	0.04187853
754477358	population	-0.001059859	6	-0.002215038	-0.0004052564	0.001119635
754477358	stimulus_time	-0.002635785	6	-0.009769743	-0.003202393	0.004160863
754477358	combined	-0.08816057	6	-0.05601582	-0.03057356	0.04357432
759711152	own	-0.02520551	6	-0.02008462	0.006575658	0.01601047
759711152	population	-0.0009553764	6	-0.0006858396	0.001079542	0.001125334
759711152	stimulus_time	-3.950937e-05	6	0.0003580485	0.004804122	0.002677252
759711152	combined	-0.0269917	6	-0.01802458	0.00350959	0.02243947
760938797	own	-0.03587188	6	-0.06233279	-0.03846997	0.01535581
760938797	population	-0.003905501	6	-0.002429006	-0.001181182	0.00153568
760938797	stimulus_time	-0.002227937	6	-0.001471055	0.0003263066	0.001850316
760938797	combined	-0.04007233	6	-0.06164029	-0.0406208	0.01185777
759674770	own	0.03048895	6	-0.05335282	0.05771702	0.07271437
759674770	population	-0.009751161	6	0.0005368766	0.002888768	0.01165412
759674770	stimulus_time	-0.004262209	6	-0.01094949	0.006174096	0.0108261
759674770	combined	0.003099253	6	-0.04500601	0.07364582	0.07203269

Mouse	Contrast	Shared-probe effect	LOPO views	LOPO min	LOPO max	Max absolute shift
760960653	own	0.008906332	6	-0.1198398	0.01668552	0.09767772
760960653	population	-0.01037886	6	-0.0106424	-0.005097794	0.003697899
760960653	stimulus_time	-0.003747744	6	-0.02200419	-0.002392547	0.01426419
760960653	combined	-0.04405326	6	-0.1492983	-0.02917393	0.08550297
760946813	own	-0.02549075	6	-0.01560945	0.001649762	0.02252386
760946813	population	-0.0006887466	6	-0.002577329	-0.001698975	0.001343164
760946813	stimulus_time	-0.001694887	6	-0.004405663	-0.002397556	0.001594334
760946813	combined	-0.0297396	6	-0.02794889	-0.01429291	0.01251752
763884103	own	-0.05339637	5	-0.04810463	-0.02929245	0.01246989
763884103	population	0.001418068	5	-0.003594107	-0.0003154037	0.003612033
763884103	stimulus_time	-0.007200976	5	-0.005396391	5.369787e-06	0.003697142
763884103	combined	-0.09020428	5	-0.06865137	-0.0332403	0.03080331
763236014	own	0.00644447	5	0.01034065	0.01579439	0.005015206
763236014	population	-0.0003359776	5	-0.002356136	0.001380143	0.002069232
763236014	stimulus_time	0.0006129672	5	-0.0003447934	0.002093034	0.001947838
763236014	combined	0.003764048	5	-0.0003128732	0.01169972	0.006183771
774672366	own	None	5	0.002677542	0.0131871	0.007903935
774672366	population	None	5	-0.01721835	-0.01042003	0.004033527
774672366	stimulus_time	None	5	-0.0101553	-0.006176919	0.002312373
774672366	combined	None	5	-0.08880657	-0.04432025	0.02670844
791857608	own	-0.01591026	5	0.0107402	0.02954344	0.03362652
791857608	population	-0.004290628	5	-0.009931554	-0.008062957	0.004737496
791857608	stimulus_time	-0.001346303	5	-0.001991619	-0.0002243543	0.001010739
791857608	combined	-0.03079777	5	-0.02187664	-0.006309101	0.01563434
795770036	own	-0.03636837	5	-0.03036813	-0.01350878	0.01595101
795770036	population	-0.004924068	5	-0.003931098	-0.0006321454	0.002227379
795770036	stimulus_time	-0.001545607	5	-0.0004703248	0.0003908591	0.001658163
795770036	combined	-0.03877374	5	-0.03176909	-0.01553134	0.01670694
800250057	own	0.00498954	3	0.04513018	0.08792441	0.0556489
800250057	population	-0.0001144776	3	-0.00539576	2.061834e-05	0.002852274
800250057	stimulus_time	0.0004978995	3	-0.0004896117	0.0008048013	0.0009144422
800250057	combined	-0.005315468	3	0.02439114	0.08130169	0.05068243
821469666	own	0.04122751	4	0.01799516	0.03928592	0.01397982
821469666	population	0.00444542	4	0.001298843	0.003225836	0.00253056
821469666	stimulus_time	0.002309982	4	-0.001144019	0.001627932	0.002097964
821469666	combined	0.04151541	4	0.01515479	0.04072616	0.0173068

All available shared/leave-probe effects, including views without enough mice for group inference: all_mouse_probe_assessment.csv and saved probe_assessment.json. Missing strata are reported as None, not imputed.

C-only rate matching residuals

Mouse	Matched pairs	Median absolute log ratio	Maximum absolute log ratio	Within log(2)
754477358	27	0.1101253	0.6821581	Yes
759711152	26	0.03905211	0.5295178	Yes
760938797	28	0.1156496	0.6699801	Yes
759674770	21	0.09729891	0.6839981	Yes
760960653	27	0.193379	0.6544263	Yes
760946813	27	0.07743445	0.4997412	Yes
763884103	28	0.1800638	0.6884633	Yes
763236014	26	0.09568973	0.5910108	Yes
774672366	29	0.09381876	0.6757554	Yes
791857608	30	0.1084884	0.5710265	Yes
795770036	23	0.1148929	0.6681294	Yes
800250057	25	0.05854242	0.6520956	Yes
821469666	25	0.08713038	0.6850674	Yes

Every matched pair log-ratio is retained in all_C_rate_matching.csv. These are calibration rates only, not D-derived QC or rate-based exclusions.

A7 All four falsification suites

Suite	Contrast	Observed pooled	Null pooled mean	Pooled Holm	Observed CA1-DG	Regional Holm
T	own	0.03508747	0.02644357	0.016	-0.00922685	0.016
T	population	0.003188255	0.002050154	0.016	-0.002952912	0.016
T	stimulus_time	0.002427168	0.004075867	1	-0.003072108	0.843
T	combined	0.04686002	0.036752	0.016	-0.02356802	0.016
S	own	0.03508747	0.03600536	1	-0.00922685	0.976
S	population	0.003188255	0.003203569	1	-0.002952912	0.976
S	stimulus_time	0.002427168	0.00161045	0.016	-0.003072108	0.09
S	combined	0.04686002	0.0460433	0.016	-0.02356802	0.09
B	own	0.03508747	0.03145869	0.016	-0.00922685	0.016
B	population	0.003188255	0.005866771	1	-0.002952912	0.016
B	stimulus_time	0.002427168	-0.008895971	0.016	-0.003072108	0.016
B	combined	0.04686002	0.03553688	0.016	-0.02356802	0.016
O	own	0.03508747	0.05877958	1	-0.00922685	0.016
O	population	0.003188255	0.004134664	1	-0.002952912	0.016
O	stimulus_time	0.002427168	0.003349915	1	-0.003072108	0.016

Suite	Contrast	Observed pooled	Null pooled mean	Pooled Holm	Observed CA1-DG	Regional Holm
O	combined	0.04686002	0.0725474	1	-0.02356802	0.016

Mouse	Region	Suite	Contrast	Observed	Null mean
754477358	CA1	B	own	0.01092217	0.01012709
754477358	CA1	B	population	0.001710043	0.001731088
754477358	CA1	B	stimulus_time	0.001830216	-0.003522973
754477358	CA1	B	combined	0.01926348	0.01391029
754477358	CA1	O	own	0.01092217	0.0179471
754477358	CA1	O	population	0.001710043	0.002014888
754477358	CA1	O	stimulus_time	0.001830216	0.001944711
754477358	CA1	O	combined	0.01926348	0.02680447
754477358	CA1	S	own	0.01092217	0.01130179
754477358	CA1	S	population	0.001710043	0.001721765
754477358	CA1	S	stimulus_time	0.001830216	0.0008261422
754477358	CA1	S	combined	0.01926348	0.01825941
754477358	CA1	T	own	0.01092217	0.009977621
754477358	CA1	T	population	0.001710043	0.0003095282
754477358	CA1	T	stimulus_time	0.001830216	0.002666426
754477358	CA1	T	combined	0.01926348	0.01513381
754477358	DG	B	own	0.04183854	0.04069333
754477358	DG	B	population	0.002805447	0.004403018
754477358	DG	B	stimulus_time	0.007439096	-0.00966394
754477358	DG	B	combined	0.06384973	0.04674669
754477358	DG	O	own	0.04183854	0.05693987
754477358	DG	O	population	0.002805447	0.003438818
754477358	DG	O	stimulus_time	0.007439096	0.009096783
754477358	DG	O	combined	0.06384973	0.08026677
754477358	DG	S	own	0.04183854	0.04327725
754477358	DG	S	population	0.002805447	0.002887272
754477358	DG	S	stimulus_time	0.007439096	0.005544052
754477358	DG	S	combined	0.06384973	0.06195469
754477358	DG	T	own	0.04183854	0.01302811
754477358	DG	T	population	0.002805447	0.002297451
754477358	DG	T	stimulus_time	0.007439096	0.007470793
754477358	DG	T	combined	0.06384973	0.02871678
759674770	CA1	B	own	0.06124223	0.05450051
759674770	CA1	B	population	0.003576719	0.001247551
759674770	CA1	B	stimulus_time	0.007108676	-0.02302593

Mouse	Region	Suite	Contrast	Observed	Null mean
759674770	CA1	B	combined	0.0821226	0.051988
759674770	CA1	O	own	0.06124223	0.1163736
759674770	CA1	O	population	0.003576719	0.009525253
759674770	CA1	O	stimulus_time	0.007108676	0.01354925
759674770	CA1	O	combined	0.0821226	0.1545704
759674770	CA1	S	own	0.06124223	0.06200987
759674770	CA1	S	population	0.003576719	0.00344278
759674770	CA1	S	stimulus_time	0.007108676	0.006379912
759674770	CA1	S	combined	0.0821226	0.08139384
759674770	CA1	T	own	0.06124223	0.05845153
759674770	CA1	T	population	0.003576719	0.003720301
759674770	CA1	T	stimulus_time	0.007108676	0.009991127
759674770	CA1	T	combined	0.0821226	0.08290908
759674770	DG	B	own	0.07623958	0.04237726
759674770	DG	B	population	0.001673762	0.008045889
759674770	DG	B	stimulus_time	0.01176068	-0.03328068
759674770	DG	B	combined	0.08050947	0.0354681
759674770	DG	O	own	0.07623958	0.1493498
759674770	DG	O	population	0.001673762	0.001861387
759674770	DG	O	stimulus_time	0.01176068	0.01289226
759674770	DG	O	combined	0.08050947	0.1557111
759674770	DG	S	own	0.07623958	0.07709422
759674770	DG	S	population	0.001673762	0.001720817
759674770	DG	S	stimulus_time	0.01176068	0.01143725
759674770	DG	S	combined	0.08050947	0.08018604
759674770	DG	T	own	0.07623958	0.06955925
759674770	DG	T	population	0.001673762	0.001746916
759674770	DG	T	stimulus_time	0.01176068	0.01260156
759674770	DG	T	combined	0.08050947	0.07667744
759711152	CA1	B	own	0.02679795	0.03254354
759711152	CA1	B	population	0.002739864	0.01275034
759711152	CA1	B	stimulus_time	0.003637366	-0.01242464
759711152	CA1	B	combined	0.03956741	0.02350541
759711152	CA1	O	own	0.02679795	0.03847341
759711152	CA1	O	population	0.002739864	0.002545278
759711152	CA1	O	stimulus_time	0.003637366	0.004284992
759711152	CA1	O	combined	0.03956741	0.05182246

Mouse	Region	Suite	Contrast	Observed	Null mean
759711152	CA1	S	own	0.02679795	0.02735654
759711152	CA1	S	population	0.002739864	0.002741487
759711152	CA1	S	stimulus_time	0.003637366	0.002990758
759711152	CA1	S	combined	0.03956741	0.0389208
759711152	CA1	T	own	0.02679795	0.02233999
759711152	CA1	T	population	0.002739864	0.0005452648
759711152	CA1	T	stimulus_time	0.003637366	0.004646694
759711152	CA1	T	combined	0.03956741	0.03041611
759711152	DG	B	own	0.035993	0.03877925
759711152	DG	B	population	0.002569906	0.00469832
759711152	DG	B	stimulus_time	0.001510497	-0.006395326
759711152	DG	B	combined	0.04411964	0.03621382
759711152	DG	O	own	0.035993	0.04791293
759711152	DG	O	population	0.002569906	0.003245865
759711152	DG	O	stimulus_time	0.001510497	0.001663538
759711152	DG	O	combined	0.04411964	0.05842718
759711152	DG	S	own	0.035993	0.03809329
759711152	DG	S	population	0.002569906	0.002624189
759711152	DG	S	stimulus_time	0.001510497	0.0008242659
759711152	DG	S	combined	0.04411964	0.04343341
759711152	DG	T	own	0.035993	0.02202517
759711152	DG	T	population	0.002569906	0.0009753829
759711152	DG	T	stimulus_time	0.001510497	0.002485602
759711152	DG	T	combined	0.04411964	0.02761038
760938797	CA1	B	own	0.0158136	0.01475669
760938797	CA1	B	population	0.0002870447	0.001011324
760938797	CA1	B	stimulus_time	0.0002667109	-0.002939906
760938797	CA1	B	combined	0.01764873	0.01444211
760938797	CA1	O	own	0.0158136	0.01985809
760938797	CA1	O	population	0.0002870447	0.0003977837
760938797	CA1	O	stimulus_time	0.0002667109	0.0002976829
760938797	CA1	O	combined	0.01764873	0.02208051
760938797	CA1	S	own	0.0158136	0.01770248
760938797	CA1	S	population	0.0002870447	0.0003261664
760938797	CA1	S	stimulus_time	0.0002667109	-0.0009869647
760938797	CA1	S	combined	0.01764873	0.01639505
760938797	CA1	T	own	0.0158136	0.01169031

Mouse	Region	Suite	Contrast	Observed	Null mean
760938797	CA1	T	population	0.0002870447	-0.000141618
760938797	CA1	T	stimulus_time	0.0002667109	0.001591547
760938797	CA1	T	combined	0.01764873	0.01471044
760938797	DG	B	own	0.06704129	0.0641041
760938797	DG	B	population	0.002656865	0.008728906
760938797	DG	B	stimulus_time	0.0006443321	-0.005842166
760938797	DG	B	combined	0.06957883	0.06309233
760938797	DG	O	own	0.06704129	0.1053979
760938797	DG	O	population	0.002656865	0.002065411
760938797	DG	O	stimulus_time	0.0006443321	0.004004396
760938797	DG	O	combined	0.06957883	0.1092993
760938797	DG	S	own	0.06704129	0.06878392
760938797	DG	S	population	0.002656865	0.002695056
760938797	DG	S	stimulus_time	0.0006443321	-0.0004263267
760938797	DG	S	combined	0.06957883	0.06850817
760938797	DG	T	own	0.06704129	0.04691867
760938797	DG	T	population	0.002656865	0.0002805024
760938797	DG	T	stimulus_time	0.0006443321	0.001860203
760938797	DG	T	combined	0.06957883	0.04839405
760946813	CA1	B	own	0.0148761	0.01525755
760946813	CA1	B	population	0.001324475	0.001175381
760946813	CA1	B	stimulus_time	-0.0005803034	-0.003249888
760946813	CA1	B	combined	0.01744238	0.0147728
760946813	CA1	O	own	0.0148761	0.0252433
760946813	CA1	O	population	0.001324475	0.002299432
760946813	CA1	O	stimulus_time	-0.0005803034	4.12403e-05
760946813	CA1	O	combined	0.01744238	0.03132918
760946813	CA1	S	own	0.0148761	0.01494506
760946813	CA1	S	population	0.001324475	0.001313689
760946813	CA1	S	stimulus_time	-0.0005803034	-0.0007096442
760946813	CA1	S	combined	0.01744238	0.01731304
760946813	CA1	T	own	0.0148761	0.01204373
760946813	CA1	T	population	0.001324475	0.0009871047
760946813	CA1	T	stimulus_time	-0.0005803034	0.001356253
760946813	CA1	T	combined	0.01744238	0.01635767
760946813	DG	B	own	0.01784299	0.01568494
760946813	DG	B	population	0.003356386	0.001259303

Mouse	Region	Suite	Contrast	Observed	Null mean
760946813	DG	B	stimulus_time	0.002708918	-0.00606683
760946813	DG	B	combined	0.03466447	0.02588872
760946813	DG	O	own	0.01784299	0.02633992
760946813	DG	O	population	0.003356386	0.002640211
760946813	DG	O	stimulus_time	0.002708918	0.002980987
760946813	DG	O	combined	0.03466447	0.04310073
760946813	DG	S	own	0.01784299	0.01778518
760946813	DG	S	population	0.003356386	0.003320654
760946813	DG	S	stimulus_time	0.002708918	0.002651795
760946813	DG	S	combined	0.03466447	0.03460734
760946813	DG	T	own	0.01784299	0.01614104
760946813	DG	T	population	0.003356386	0.004624833
760946813	DG	T	stimulus_time	0.002708918	0.004204898
760946813	DG	T	combined	0.03466447	0.03481773
760960653	CA1	B	own	0.02746109	0.02722265
760960653	CA1	B	population	0.001436633	0.001200013
760960653	CA1	B	stimulus_time	-0.0006705078	-0.003067986
760960653	CA1	B	combined	0.02946726	0.02706978
760960653	CA1	O	own	0.02746109	0.0380183
760960653	CA1	O	population	0.001436633	0.001808716
760960653	CA1	O	stimulus_time	-0.0006705078	-0.0006267753
760960653	CA1	O	combined	0.02946726	0.04066704
760960653	CA1	S	own	0.02746109	0.02753364
760960653	CA1	S	population	0.001436633	0.001453412
760960653	CA1	S	stimulus_time	-0.0006705078	-0.0009464669
760960653	CA1	S	combined	0.02946726	0.0291913
760960653	CA1	T	own	0.02746109	0.0222576
760960653	CA1	T	population	0.001436633	-0.000210665
760960653	CA1	T	stimulus_time	-0.0006705078	0.001076699
760960653	CA1	T	combined	0.02946726	0.02337019
760960653	DG	B	own	0.1084533	0.04997513
760960653	DG	B	population	0.008381136	0.01490516
760960653	DG	B	stimulus_time	0.01598623	-0.04957793
760960653	DG	B	combined	0.1441442	0.07857999
760960653	DG	O	own	0.1084533	0.2188411
760960653	DG	O	population	0.008381136	0.006655942
760960653	DG	O	stimulus_time	0.01598623	0.02371214

Mouse	Region	Suite	Contrast	Observed	Null mean
760960653	DG	O	combined	0.1441442	0.2556795
760960653	DG	S	own	0.1084533	0.1087651
760960653	DG	S	population	0.008381136	0.008397107
760960653	DG	S	stimulus_time	0.01598623	0.01585997
760960653	DG	S	combined	0.1441442	0.1440179
760960653	DG	T	own	0.1084533	0.09308228
760960653	DG	T	population	0.008381136	0.008266786
760960653	DG	T	stimulus_time	0.01598623	0.01788525
760960653	DG	T	combined	0.1441442	0.1197922
763236014	CA1	B	own	0.02503814	0.02499892
763236014	CA1	B	population	0.001920921	0.002343647
763236014	CA1	B	stimulus_time	6.581798e-05	-0.003959682
763236014	CA1	B	combined	0.02894393	0.02491843
763236014	CA1	O	own	0.02503814	0.0350352
763236014	CA1	O	population	0.001920921	0.002370299
763236014	CA1	O	stimulus_time	6.581798e-05	0.0002531575
763236014	CA1	O	combined	0.02894393	0.04037627
763236014	CA1	S	own	0.02503814	0.02562073
763236014	CA1	S	population	0.001920921	0.001922057
763236014	CA1	S	stimulus_time	6.581798e-05	-0.000311854
763236014	CA1	S	combined	0.02894393	0.02856626
763236014	CA1	T	own	0.02503814	0.01940588
763236014	CA1	T	population	0.001920921	-0.0001401969
763236014	CA1	T	stimulus_time	6.581798e-05	0.001630756
763236014	CA1	T	combined	0.02894393	0.02083879
763236014	DG	B	own	0.01425896	0.01400782
763236014	DG	B	population	0.002207824	0.001506744
763236014	DG	B	stimulus_time	-7.937811e-05	-0.003297027
763236014	DG	B	combined	0.02307303	0.01985538
763236014	DG	O	own	0.01425896	0.02195857
763236014	DG	O	population	0.002207824	0.003092295
763236014	DG	O	stimulus_time	-7.937811e-05	0.0001277877
763236014	DG	O	combined	0.02307303	0.03609274
763236014	DG	S	own	0.01425896	0.01414266
763236014	DG	S	population	0.002207824	0.002232389
763236014	DG	S	stimulus_time	-7.937811e-05	-0.0002342805
763236014	DG	S	combined	0.02307303	0.02291813

Mouse	Region	Suite	Contrast	Observed	Null mean
763236014	DG	T	own	0.01425896	0.01022268
763236014	DG	T	population	0.002207824	0.00175609
763236014	DG	T	stimulus_time	-7.937811e-05	0.002291316
763236014	DG	T	combined	0.02307303	0.01768291
763884103	CA1	B	own	0.01740096	0.0160308
763884103	CA1	B	population	0.001595902	0.005631967
763884103	CA1	B	stimulus_time	0.001043259	-0.002893681
763884103	CA1	B	combined	0.02331541	0.01937847
763884103	CA1	O	own	0.01740096	0.03991127
763884103	CA1	O	population	0.001595902	0.003620792
763884103	CA1	O	stimulus_time	0.001043259	-7.734737e-05
763884103	CA1	O	combined	0.02331541	0.04612935
763884103	CA1	S	own	0.01740096	0.02335727
763884103	CA1	S	population	0.001595902	0.001855416
763884103	CA1	S	stimulus_time	0.001043259	-0.002280227
763884103	CA1	S	combined	0.02331541	0.01999192
763884103	CA1	T	own	0.01740096	0.009324995
763884103	CA1	T	population	0.001595902	-0.0008123522
763884103	CA1	T	stimulus_time	0.001043259	0.002317486
763884103	CA1	T	combined	0.02331541	0.01493167
763884103	DG	B	own	0.05832744	0.07662443
763884103	DG	B	population	0.003789867	0.00858507
763884103	DG	B	stimulus_time	0.004547093	-0.01071375
763884103	DG	B	combined	0.08271638	0.06745553
763884103	DG	O	own	0.05832744	0.09030459
763884103	DG	O	population	0.003789867	0.005241947
763884103	DG	O	stimulus_time	0.004547093	0.005038863
763884103	DG	O	combined	0.08271638	0.1163018
763884103	DG	S	own	0.05832744	0.06395191
763884103	DG	S	population	0.003789867	0.003752626
763884103	DG	S	stimulus_time	0.004547093	-0.001970367
763884103	DG	S	combined	0.08271638	0.07619892
763884103	DG	T	own	0.05832744	0.0170509
763884103	DG	T	population	0.003789867	-0.001065479
763884103	DG	T	stimulus_time	0.004547093	0.006581123
763884103	DG	T	combined	0.08271638	0.03767909
774672366	CA1	B	own	0.02047243	0.01965068

Mouse	Region	Suite	Contrast	Observed	Null mean
774672366	CA1	B	population	0.0005664499	0.001939162
774672366	CA1	B	stimulus_time	-0.0008131872	-0.003019225
774672366	CA1	B	combined	0.0203823	0.01817626
774672366	CA1	O	own	0.02047243	0.02502936
774672366	CA1	O	population	0.0005664499	0.0006665405
774672366	CA1	O	stimulus_time	-0.0008131872	-0.0007817645
774672366	CA1	O	combined	0.0203823	0.02493717
774672366	CA1	S	own	0.02047243	0.02070855
774672366	CA1	S	population	0.0005664499	0.0005935053
774672366	CA1	S	stimulus_time	-0.0008131872	-0.0009317383
774672366	CA1	S	combined	0.0203823	0.02026375
774672366	CA1	T	own	0.02047243	0.0180687
774672366	CA1	T	population	0.0005664499	7.147263e-05
774672366	CA1	T	stimulus_time	-0.0008131872	0.001170211
774672366	CA1	T	combined	0.0203823	0.01904104
774672366	DG	B	own	0.009890954	0.01043729
774672366	DG	B	population	0.01375128	0.008475743
774672366	DG	B	stimulus_time	0.007029745	-0.01341337
774672366	DG	B	combined	0.08248043	0.06203731
774672366	DG	O	own	0.009890954	0.01629313
774672366	DG	O	population	0.01375128	0.01251167
774672366	DG	O	stimulus_time	0.007029745	0.007008565
774672366	DG	O	combined	0.08248043	0.09512874
774672366	DG	S	own	0.009890954	0.01001777
774672366	DG	S	population	0.01375128	0.01372974
774672366	DG	S	stimulus_time	0.007029745	0.006753059
774672366	DG	S	combined	0.08248043	0.08220374
774672366	DG	T	own	0.009890954	0.005242133
774672366	DG	T	population	0.01375128	0.01730646
774672366	DG	T	stimulus_time	0.007029745	0.01023036
774672366	DG	T	combined	0.08248043	0.07936128
791857608	CA1	B	own	0.02788284	0.02545196
791857608	CA1	B	population	0.001393532	0.004382754
791857608	CA1	B	stimulus_time	-0.0005002832	-0.003861816
791857608	CA1	B	combined	0.0268824	0.02352087
791857608	CA1	O	own	0.02788284	0.04884169
791857608	CA1	O	population	0.001393532	0.002975458

Mouse	Region	Suite	Contrast	Observed	Null mean
791857608	CA1	O	stimulus_time	-0.0005002832	-0.0002426924
791857608	CA1	O	combined	0.0268824	0.04860923
791857608	CA1	S	own	0.02788284	0.02753267
791857608	CA1	S	population	0.001393532	0.001444207
791857608	CA1	S	stimulus_time	-0.0005002832	-0.0006368712
791857608	CA1	S	combined	0.0268824	0.02674581
791857608	CA1	T	own	0.02788284	0.02747259
791857608	CA1	T	population	0.001393532	0.0002454831
791857608	CA1	T	stimulus_time	-0.0005002832	0.001018996
791857608	CA1	T	combined	0.0268824	0.02679921
791857608	DG	B	own	0.01016658	0.009874135
791857608	DG	B	population	0.01042166	0.009260742
791857608	DG	B	stimulus_time	0.0007348096	-0.004177648
791857608	DG	B	combined	0.04204583	0.03713337
791857608	DG	O	own	0.01016658	0.01235229
791857608	DG	O	population	0.01042166	0.009725976
791857608	DG	O	stimulus_time	0.0007348096	0.0008619649
791857608	DG	O	combined	0.04204583	0.04389145
791857608	DG	S	own	0.01016658	0.01025343
791857608	DG	S	population	0.01042166	0.0103363
791857608	DG	S	stimulus_time	0.0007348096	0.0007332408
791857608	DG	S	combined	0.04204583	0.04204426
791857608	DG	T	own	0.01016658	0.006295032
791857608	DG	T	population	0.01042166	0.007330676
791857608	DG	T	stimulus_time	0.0007348096	0.003670352
791857608	DG	T	combined	0.04204583	0.02896013
795770036	CA1	B	own	0.01612762	0.0128904
795770036	CA1	B	population	0.0007015575	0.00766197
795770036	CA1	B	stimulus_time	0.0005348186	-0.004921251
795770036	CA1	B	combined	0.01690577	0.0114497
795770036	CA1	O	own	0.01612762	0.02890527
795770036	CA1	O	population	0.0007015575	0.0008977795
795770036	CA1	O	stimulus_time	0.0005348186	0.0005848432
795770036	CA1	O	combined	0.01690577	0.02974456
795770036	CA1	S	own	0.01612762	0.01675775
795770036	CA1	S	population	0.0007015575	0.0007660242
795770036	CA1	S	stimulus_time	0.0005348186	-0.0002159062

Mouse	Region	Suite	Contrast	Observed	Null mean
795770036	CA1	S	combined	0.01690577	0.01615505
795770036	CA1	T	own	0.01612762	0.0108266
795770036	CA1	T	population	0.0007015575	8.010975e-05
795770036	CA1	T	stimulus_time	0.0005348186	0.001656561
795770036	CA1	T	combined	0.01690577	0.01233815
795770036	DG	B	own	0.03654498	0.03449633
795770036	DG	B	population	0.003398246	0.01685855
795770036	DG	B	stimulus_time	0.0004222629	-0.005274687
795770036	DG	B	combined	0.03897256	0.03327561
795770036	DG	O	own	0.03654498	0.1017449
795770036	DG	O	population	0.003398246	0.01298997
795770036	DG	O	stimulus_time	0.0004222629	0.0009599808
795770036	DG	O	combined	0.03897256	0.1053419
795770036	DG	S	own	0.03654498	0.03706649
795770036	DG	S	population	0.003398246	0.00343197
795770036	DG	S	stimulus_time	0.0004222629	-0.0002288431
795770036	DG	S	combined	0.03897256	0.03832146
795770036	DG	T	own	0.03654498	0.02477281
795770036	DG	T	population	0.003398246	0.0006423806
795770036	DG	T	stimulus_time	0.0004222629	0.001242632
795770036	DG	T	combined	0.03897256	0.0257714
800250057	CA1	B	own	0.07819137	0.07556386
800250057	CA1	B	population	0.001579337	0.004437766
800250057	CA1	B	stimulus_time	-0.0004838941	-0.005945944
800250057	CA1	B	combined	0.07784705	0.072385
800250057	CA1	O	own	0.07819137	0.08953906
800250057	CA1	O	population	0.001579337	0.001704835
800250057	CA1	O	stimulus_time	-0.0004838941	-0.0005397447
800250057	CA1	O	combined	0.07784705	0.08962872
800250057	CA1	S	own	0.07819137	0.07740577
800250057	CA1	S	population	0.001579337	0.001530954
800250057	CA1	S	stimulus_time	-0.0004838941	-0.000228377
800250057	CA1	S	combined	0.07784705	0.07810257
800250057	CA1	T	own	0.07819137	0.07035855
800250057	CA1	T	population	0.001579337	0.00043702
800250057	CA1	T	stimulus_time	-0.0004838941	0.001507508
800250057	CA1	T	combined	0.07784705	0.06646254

Mouse	Region	Suite	Contrast	Observed	Null mean
800250057	DG	B	own	0.01755293	0.01769158
800250057	DG	B	population	0.004122824	0.003840355
800250057	DG	B	stimulus_time	-0.0003742532	-0.002849574
800250057	DG	B	combined	0.03248009	0.03000477
800250057	DG	O	own	0.01755293	0.02127253
800250057	DG	O	population	0.004122824	0.003919546
800250057	DG	O	stimulus_time	-0.0003742532	-0.0004416148
800250057	DG	O	combined	0.03248009	0.03608332
800250057	DG	S	own	0.01755293	0.01781606
800250057	DG	S	population	0.004122824	0.004147521
800250057	DG	S	stimulus_time	-0.0003742532	-0.0005727042
800250057	DG	S	combined	0.03248009	0.03228164
800250057	DG	T	own	0.01755293	0.01034997
800250057	DG	T	population	0.004122824	0.002416335
800250057	DG	T	stimulus_time	-0.0003742532	0.001283916
800250057	DG	T	combined	0.03248009	0.02144313
821469666	CA1	B	own	0.05393602	0.05286202
821469666	CA1	B	population	0.003420913	0.01358099
821469666	CA1	B	stimulus_time	0.0001457983	-0.005446243
821469666	CA1	B	combined	0.05619934	0.0506073
821469666	CA1	O	own	0.05393602	0.1101455
821469666	CA1	O	population	0.003420913	0.007679359
821469666	CA1	O	stimulus_time	0.0001457983	0.001392533
821469666	CA1	O	combined	0.05619934	0.1161032
821469666	CA1	S	own	0.05393602	0.05459829
821469666	CA1	S	population	0.003420913	0.003406179
821469666	CA1	S	stimulus_time	0.0001457983	-0.0003692684
821469666	CA1	S	combined	0.05619934	0.05568428
821469666	CA1	T	own	0.05393602	0.0416178
821469666	CA1	T	population	0.003420913	0.001120547
821469666	CA1	T	stimulus_time	0.0001457983	0.00190755
821469666	CA1	T	combined	0.05619934	0.04282949
821469666	DG	B	own	0.02196104	0.02132367
821469666	DG	B	population	0.001506053	0.00287431
821469666	DG	B	stimulus_time	-0.0008081462	-0.002463136
821469666	DG	B	combined	0.02373776	0.02208277
821469666	DG	O	own	0.02196104	0.02624036

Mouse	Region	Suite	Contrast	Observed	Null mean
821469666	DG	O	population	0.001506053	0.001605806
821469666	DG	O	stimulus_time	-0.0008081462	-0.0008879553
821469666	DG	O	combined	0.02373776	0.0281055
821469666	DG	S	own	0.02196104	0.02226171
821469666	DG	S	population	0.001506053	0.0014995
821469666	DG	S	stimulus_time	-0.0008081462	-0.001078895
821469666	DG	S	combined	0.02373776	0.02346701
821469666	DG	T	own	0.02196104	0.01900888
821469666	DG	T	population	0.001506053	0.0005136811
821469666	DG	T	stimulus_time	-0.0008081462	0.001626713
821469666	DG	T	combined	0.02373776	0.0225072

All104 complete999x4 replicate arrays remain in saved results. No null fitting, selected suite replacement or trial repartitioning. These conditional reference distributions have nuisance/exchangeability assumptions.

A8 QC and leakage audit

The exact immutable C bundle and failed partial bundle were hash-checked; the original complete C provenance/numerical validator passed. C-only QC, unit/scaler/trial selections and source masks were not recalculated using D. All required views retain13 mice; no D-derived QC, normalization or exclusions. Synthetic transport tests used artificial counts, authenticated envelopes, exact masks, ordered-unit canaries, tamper/wrong-key negatives, and no real source. Existing exposure records are evidence of recorded isolation, not proof against unlogged administrator activity. Whole-session upstream sorting remains a limitation.

A9 Failures, limitations and claim audit

Contrast	Registered classification	Holm criterion met	CI excludes zero	Development direction	Sensitivity signs	All LOMO signs	Relevant corrected control
Own history	NOT VALIDATED / MIXED	No	No	No	No	No	Yes
Other-unit history	NOT VALIDATED / MIXED	No	Yes	Yes	Yes	Yes	Yes
Stimulus/time	NOT VALIDATED / MIXED	No	Yes	Yes	Yes	Yes	Yes
Combined features	NOT VALIDATED / MIXED	No	Yes	Yes	Yes	Yes	Yes

All original scientific M4 and prerelease failures, finite-difference diagnosis, three nonconvergent conditioning candidates, partial seal/import failure and later repairs remain immutable. This mission does not retroactively validate them. Development is exploratory, anatomy is provider-proxy based, selected-unit populations and rate residuals remain confounds, and controls are conditional. Scientific claim status follows every original gate, with no equivalence/causal/human/NAP/CWP/independent-laboratory claim. Author declarations require owner review before submission.

A10 Saved-only reproduction

From repository root:

```
& ./paper12_sciencefinal1/reproduce_all.ps1
```

This runs synthetic numerical/transport regressions, verifies final scalar-result hashes, reaggregates all primary/view statistics and both complete null families through the frozen implementation, and regenerates manuscript/figures/PDFs in a fresh ignored work directory. It performs zero fits and opens no raw counts, C states or D key/cipher. It is not permission or a method for repeating the one-shot D assessment. Original execution provenance is certified by saved preflight/run receipts; environment identity is recorded separately. Core scalar-only checks are also available through reproduce.ps1. Artifact generation requires the recorded Matplotlib environment, bundled ReportLab/PyPDF runtime and Windows Arial fonts; missing dependencies fail explicitly, without synthetic substitution.